EVOBI AUTOMATIONS PVT. LTD.

Process Observation & Improvement Proposal

Objective

The objective of this study was to observe the SMT PCB assembly and inspection process, identify recurring quality-related issues across different production lines, and propose practical improvements to enhance inspection efficiency, defect traceability, and process consistency.

Process Observations

During the observation of SMT assembly, THT assembly, depaneling, and quality inspection activities, the following issues were identified:

• QC inspectors occasionally perform minor rework activities such as cleaning flux residue and resoldering solder joints during inspection. This increases inspection time and reduces inspection efficiency.

• The existing A, B and C grading system indicates only defect severity but does not specify the actual defect type, making root cause analysis difficult.

• Defects are not recorded using a standardized coding system, limiting trend analysis, recurring defect identification and corrective action planning.

• Manual depaneling and PCB edge scraping occasionally result in over-scraping or under-scraping, causing edge damage.

• Variations in bare PCB quality from suppliers occasionally result in copper exposure, scratches or surface damage, producing non-repairable boards.

Proposed Defect Classification

Code	Category	Defect	Description
D1	PCB / Bare Board	**Bare Board Defect**	Copper exposed, damaged PCB, manufacturing defects
D2	Component Placement	**Missing Component**	Component not mounted
D3	Component Placement (SMT)	**Misalignment**	Component shifted or rotated
D4	Component Placement (SMT)	**Polarity Error**	Incorrect component polarity
D5	SMT Soldering	**Improper SMT Solder**	Insufficient, excessive or defective solder joint
D6	THT Soldering	**Improper THT Solder**	Poor solder cone or insufficient through-hole solder
D7	Thermal Damage	**Component / PCB Burnt**	Heat damage to PCB or electronic component
D8	Mechanical (Depaneling)	**Over Scraping**	Excessive PCB edge scraping
D9	Mechanical (Depaneling)	**Under Scraping**	Incomplete PCB edge finishing
D10	Surface Quality	**Flux Residue**	Flux not completely cleaned after soldering
D11	Surface Quality	**Surface Scratch**	Visible scratches or cosmetic PCB damage

Conclusion

This proposal introduces a standardized defect classification method that can improve inspection consistency, defect tracking, and quality analysis within the SMT manufacturing process. Implementing a defect coding system can support data-driven decision making, facilitate continuous improvement, and contribute to higher overall manufacturing quality.

Prepared as a process observation and improvement proposal based on SMT manufacturing activities performed during internship at EVOBI AUTOMATIONS PVT. LTD.